

PROJECT 2

SECOND ROBOTICS PROJECT

ROBOTICS



POLITECNICO
MILANO 1863

→ recorded file of this MOBILE ROBOT :



THE PROBLEM

↓ TASKS:

- Map creation
- Autonomous navigation
with
- Multiple sensors
- Skid steering robot, with footprint: 0.6m x 0.4m



DATA

Bag File given: What is inside (map inside the Lab)
(simpler than FIELD)



Format: ROS Bag files

play the bags with the command:

roslaunch play --clock first.bag

inside LAB
lot, feature, used to
create map

⇓ What data:

Data:

- /scan: data from a single plane scanner (laser scanner single plane)
- /t265/odom: odometry data (visual odometry, multiple sensor)
↓ odometry data, easier
than encoder data
(SKID STEERING HARD!)
↳ better visual odom
- /velodyne_points: data from 3d laser
↑ 3D information from 3D laser mounted
(NOT so "grains" → dense)



THE PROJECT

- Create a ROS package called **second_project** → takes odometry and publish it as Tf
 - Create a ROS node to convert the odometry topic into tf:
 - node called **tf_publisher** → for navigation you need a tf between odom..
 - Convert the 3D lidar data to a 2D laser scanner:
 - ↓ map of environment, one from 2D laser (single plane), one using the converted 3D data (should be similar)
 - Create a map of the environment with 2D laser data (use 2D Data)
 - Create a map of the environment with 3D laser data, converted to 2D
(to fix the height where to cut: single plane scanner → take 3D Data, cut it at decide the once Best for you) ← fixed height
- after SET UP.. ↓
- Use the map to setup a "realistic" stage simulation (similar size/fov)
 - Navigate the simulation (as the one seen): you can use the map build using the tool when mapping

"realistic": similar size of Robot and field of view (fov) of laser scanner
(leave noise to Q → don't simulate odometry ERROR)



THE PROJECT

using the stage set up...

- Navigate the simulation:
 - node called **navigation**
 - write a node that given a set of waypoint from a **csv file** read them and **publish the first goal**, movebase handle the movement to it, when reached the next goal is published, etc.
- If additional nodes are needed you can add them

publish the goal to the different waypoints
(so create waypoints, publish goal, robot moves
given feedback until reached goal, then you can give
2nd goal)

(Robot navigate towards you)
{ write your own waypoint
for testing! }

Allowed minimal manual post-processing of the computed map to use it for the simulation (remove noisy points)

← you can clean a bit the map
if problems, because it
complicate...

Provide also the original maps

THE PROJECT

how to START all!



1) Launch file to start the mapping process:

- start all the required nodes + rviz (with proper config file). Do not start bag file (these are started autonomously) ⇒ bag STARTED by prof!

robot moving around and creating the map



2) Launch file to start the autonomous navigation:

- start all the required nodes + rviz (with proper config file)

path, stage, gmap, ...

robot moving on simulation

File structure



- **second_project**
 - **cfg** (if needed)
 - **launch** → inside launch file
I should START stage! DON'T use absolute path, but the project name
 - **src**
 - **srv**
 - **map_raw** → **map** from the mapping pipeline without postprocessing
 - **stage** → stage simulation: contains all stage set-up
 - **waypoints.csv** → the list of waypoints: contains all the different waypoints where Robot will navigate
 - **config_mapping.rviz**
 - **config_nav.rviz**
- ↑ two config file of rviz
- at the end you will have 4 maps, 8 ram 2 SENSORS and 2 LAUNCH files.. → here put all maps created
- ↑ contains map created with the mapping part



Maps, waypoints



Provide all 4 reconstructed maps:

- 2 maps with single plane
 - 2 maps with multi plane :
- [decide which one to use for navigation!]

⇓ structure like this:

Waypoints

CSV

file:

pose1.x, pose1.y, heading1
pose2.x, pose2.y, heading2
...

} each line is one waypoint.
(x,y, θ) to reach the goal, convert this to quaternion and send it as goal for stage

Other info



- assume `tf 0 0 0` between odometry and lasers *sensors lays on odom all aligned on same direction*
- File `txt` must contain only the group names with this structure
`codice persona;name;surname`
- File `readme` with info on the employed node/plugins (what have you used for mapping, which local planner, which global planner, etc.)
↳ Mapping, toolbox... specify used notes



Deadlines and requested files

Deadline: 30 June (1 month)

Max 3 student for team

Questions:

- write to me via mail (simone.mentasti@polimi.it)
- do not write only to Prof. Matteucci

